# Vaibhav Sen Malla

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## Experience

**Software Engineer** **New York, NY**

*Contractor.AI* *May 2025 – December 2025*

- Built and deployed a secure contractor registry tool in TypeScript & React for contractor onboarding and management
- Designed and implemented a Supabase SQL schema enabling <200ms query latency for contractor data retrieval
- Collaborated closely with designers to implement UI/UX changes in a fast-paced agile setting
- Analyzed user behavior using heatmaps and Google Analytics to improve landing page SEO and user engagement
- Leveraged foundation models and AI-assisted development workflows to accelerate coding, debugging, and deployment

**Software Engineer II** **Boise, ID**

*Ericsson* *Dec 2021 – Apr 2024*

- Built and maintained NetCloud Exchange, a 5G-optimized SD-WAN platform for enterprise-scale, cloud-native deployments
- Led a high-priority cross-team AWS data migration project, delivering 50% ahead of schedule and refactoring 1,000+ LOC
- Revamped Makefiles and Docker containers, reducing build time and cutting test failures by 75%
- Automated ∼10 CI/CD pipelines for AWS and Azure deployments using Python, Jenkins, and Docker, reducing deployment time from 5 hours to 18 minutes (94%) across 1,000+ total runs, saving thousands of engineering hours
- Owned planning, deployment, and documentation for monthly production releases of the NetCloud Exchange

**Co-Founder & Product Manager** **University Park, PA**

*Gage Ventures* *Feb 2020 – July 2020*

- Led product design, software architecture, and user research to shape MVP features and UX
- Directed development of MVP using React Native and Node.js to test product-market fit
- Selected into Penn State's Startup Accelerator (Summer Founders Program), securing $15,000 in funding

**HCI Research Assistant** **University Park, PA**

*Penn State's College of Information Sciences & Technology* *May 2019 – May 2020*

- Contributed to the "Visual Cortex on Silicon" project using AR, computer vision, and haptics for visually impaired navigation
- Conducted usability testing of the haptic prosthetics and scenario-based user studies (e.g., grocery store navigation)
- Designed wireframes and interactive prototypes in Figma and Adobe XD for A/B testing and publications

## Projects

**IBD Risk Prediction (Applied Machine Learning)** *Dec 2025*

- Predicted rare disease (1.8% prevalence) from 62K survey records, achieving 0.749 ROC AUC using ensemble methods, feature selection, imbalance handling, and time-based validation

**Airline Network Fragility Analysis** *Dec 2025*

- Analyzed 60K+ airline routes with Python/NetworkX to model disruption propagation and identify critical hub vulnerabilities

**AI Commit Generator** *June 2025*

- CLI tool that analyzes git diffs using LLMs and generates meaningful commit messages, improving developers' Git workflow

**Wireflow** *May 2025*

- Built a full-stack web application that converts wireframes and UI screenshots into clean, responsive React/JSX code using multimodal LLMs, with automated layout detection and syntax-highlighted output

## Education

**University of Michigan** **Ann Arbor, MI**

*M.S. in Information (Data Science track) | GPA: 3.91/4.00* *Expected: May 2026*

**Pennsylvania State University** **University Park, PA**

*B.S. in Information Sciences & Technology (Software Design & Development track)* *Graduated: Dec 2020*

**Relevant Coursework & Certifications:**

Harvard's CS50x | Princeton's Algorithms II | Data Structures & Algorithms | Georgia Tech's AI, Ethics, & Society
Applied Machine Learning | Information Retrieval | Data Mining | Network Analysis | Probability & Statistics
Linear Algebra & Calculus | Discrete Math | Distributed Systems | Data Analysis & Manipulation Full-Stack Web Development

## Skills

**Languages & Frameworks**: Python, SQL, R, Java, TypeScript, JavaScript, HTML/CSS, Node.js, React, REST APIs
**AI/ML & Data**: Pandas, NumPy, Scikit-learn, NLP, Feature Engineering, Model Evaluation, Data Visualization, PostgreSQL
**Cloud & DevOps**: AWS, Azure, GCP, Docker, Jenkins, CI/CD, Linux, Bash, Git, Supabase